

Phase 6: Project Testing Phase

1. Introduction

The Project Testing Phase ensures that the **Smart Budget AI** system meets all functional, security, performance, and usability requirements. This phase verifies that the AI models, budget tracking algorithms, database synchronization, and user interfaces function accurately without errors before deployment.

2. Objectives of Testing

- Verify the accuracy of AI-driven expense categorization and budget recommendations.
- Ensure data consistency across backend services and user databases.
- Test the stability, response time, and scalability under varying user loads.
- Validate secure handling of financial data and compliance with encryption standards.

3. Testing Framework & Scope

| Test Strategy             | Tools / Frameworks                                  | Scope / Focus Area                                                        |
|---------------------------|-----------------------------------------------------|---------------------------------------------------------------------------|
| Unit Testing              | Jest, PyTest                                        | Testing individual functions (e.g., currency converter, category parser). |
| Integration Testing       | Postman, Supertest                                  | Verifying API endpoints and database communication.                       |
| AI/ML Model Testing       | Scikit-Learn evaluation metrics, Custom Test Suites | Model precision, recall, accuracy, and dataset edge cases.                |
| UI/UX (Usability) Testing | Cypress, React Testing Library                      | Navigation flow, responsiveness, accessible user interfaces.              |
| Security Testing          | OWASP ZAP, Burp Suite                               | Authentication security, JWT validation, SQL injection prevention.        |
| Performance Testing       | JMeter                                              | API latency, server load under high concurrent requests.                  |

4. Detailed Test Cases

Test Case 1: AI Expense Categorization

- **Test Case ID:** TC\_AI\_01
- **Feature:** Automatic Expense Classification
- **Input Data:** Transaction Description: "Starbucks Coffee \$5.50"
- **Expected Result:** System automatically categorizes the transaction under **"Food & Dining"**.
- **Actual Result:** Categorized under **"Food & Dining"**.
- **Status:** PASS

Test Case 2: Budget Exceeded Notification

- **Test Case ID:** TC\_NOTIF\_02
- **Feature:** Real-time Alerting
- **Input Data:** User enters an expense that exceeds the set limit for "Entertainment".
- **Expected Result:** System triggers an immediate push notification and displays a budget warning banner.
- **Actual Result:** Notification received in < 1 second with warning UI update.
- **Status:** PASS

Test Case 3: Authentication & Security

- **Test Case ID:** TC\_SEC\_03
- **Feature:** User Login & JWT Token
- **Input Data:** Login attempt with invalid credentials.
- **Expected Result:** System returns HTTP 401 Unauthorized error with clear messaging.
- **Actual Result:** Access denied; user token not generated.
- **Status:** PASS

5. Bug Report & Resolution Matrix

| Bug ID | Severity | Description                                                      | Status | Resolution                                                         |
|--------|----------|------------------------------------------------------------------|--------|--------------------------------------------------------------------|
| BUG-01 | High     | Machine learning model miscategorized "Uber Ride" as "Shopping". | Fixed  | Re-trained model with updated dataset containing transport labels. |
| BUG-02 | Medium   | Dashboard rendering delayed by 3.5s on mobile resolution.        | Fixed  | Optimized API payload size and lazy-loaded chart components.       |
| BUG-03 | Low      | Minor typographical error in budget forecast chart legend.       | Fixed  | Corrected string resources in the frontend code.                   |

6. Testing Conclusion & Deliverables

All core functionalities of **Smart Budget AI**—including automated expense logging, AI-driven insights, notification alerts, and data security—were rigorously tested. The system achieved a **98% test case pass rate**, meeting all critical baseline metrics for performance and reliability, making it ready for **Phase 7: Project Documentation Phase**.